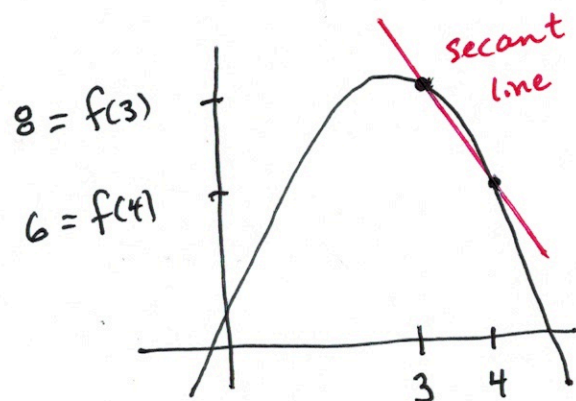


Section 2.7 Derivatives As Rates of Change

Ex ①

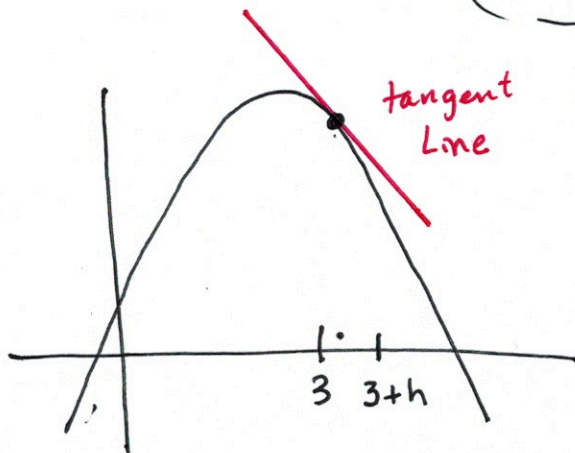


$$f(x) = -x^2 + 5x + 2$$

$$m = \frac{f(4) - f(3)}{4 - 3}$$

$$m = \frac{6 - 8}{1}$$

$$m = -2$$



Slope of tangent line

$$\lim_{h \rightarrow 0} \frac{f(3+h) - f(3)}{h}$$

$$\lim_{h \rightarrow 0} \frac{-(3+h)^2 + 5(3+h) + 2 - 8}{h}$$

Ex ① continued

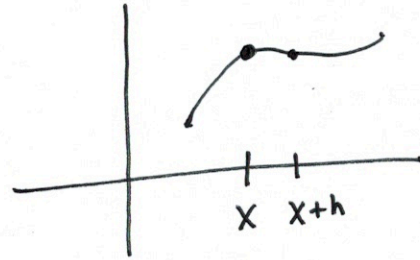
$$\lim_{h \rightarrow 0} \frac{-\cancel{9} - 6h - h^2 + \cancel{15} + h \cdot 5 - \cancel{6}}{h}$$

$$\lim_{h \rightarrow 0} \frac{-h^2 - h}{h} \rightarrow \frac{\cancel{h}(-h-1)}{\cancel{h}}$$

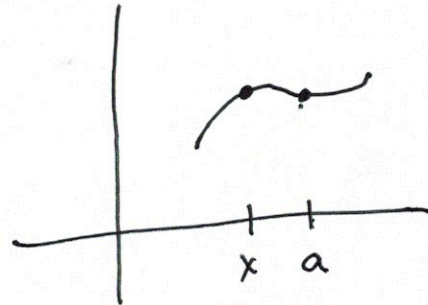
$$\lim_{h \rightarrow 0} -h - 1$$

$$= -1$$

Slope of a tangent line



$$\lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$



$$\lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a}$$

Ex ② Find slope of the tangent line
for $f(x) = \sqrt{x}$ when $a = 4$.

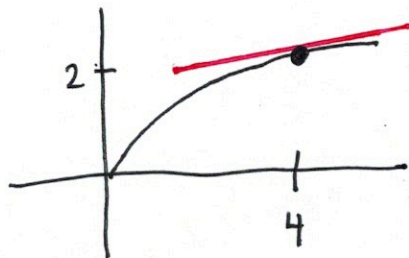
$$\lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a}$$

$$\lim_{x \rightarrow 4} \frac{\sqrt{x} - \sqrt{4}}{x - 4}$$

$$\lim_{x \rightarrow 4} \frac{(\sqrt{x} - 2)(\sqrt{x} + 2)}{(x - 4)(\sqrt{x} + 2)}$$

$$\lim_{x \rightarrow 4} \frac{\cancel{x - 4} \cdot 1}{(\cancel{x - 4})(\sqrt{x} + 2)}$$

$$\lim_{x \rightarrow 4} \frac{1}{\sqrt{x} + 2} = \frac{1}{\sqrt{4} + 2} = \boxed{\frac{1}{4}}$$



Ex ③ Drop a ball from the top of the Empire State Building.

After t seconds the height (in feet) would be $h(t) = -16t^2 + 1250$.

A) Find the average velocity over $[4, 5]$

$$\frac{h(5) - h(4)}{5 - 4} = \frac{850 - 994}{1} = -144 \frac{\text{ft}}{\text{sec}}$$

B) Find instantaneous velocity when $t = 8 \text{ sec}$:

$$\lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} = \lim_{h \rightarrow 0} \frac{f(8+h) - f(8)}{h}$$

$$\lim_{h \rightarrow 0} \frac{-16(8+h)^2 + 1250 - [-16(8)^2 + 1250]}{h}$$

$$\lim_{h \rightarrow 0} \frac{-16(64 + 16h + h^2) + 1024}{h}$$

Ex ③ cont'd

$$\lim_{h \rightarrow 0} \frac{-\cancel{1024} - 256h - 16h^2 + \cancel{1024}}{h}$$

$$\lim_{h \rightarrow 0} \frac{\cancel{h}(-256 - 16h)}{\cancel{h}}$$

$$= -256 \text{ ft/sec.}$$